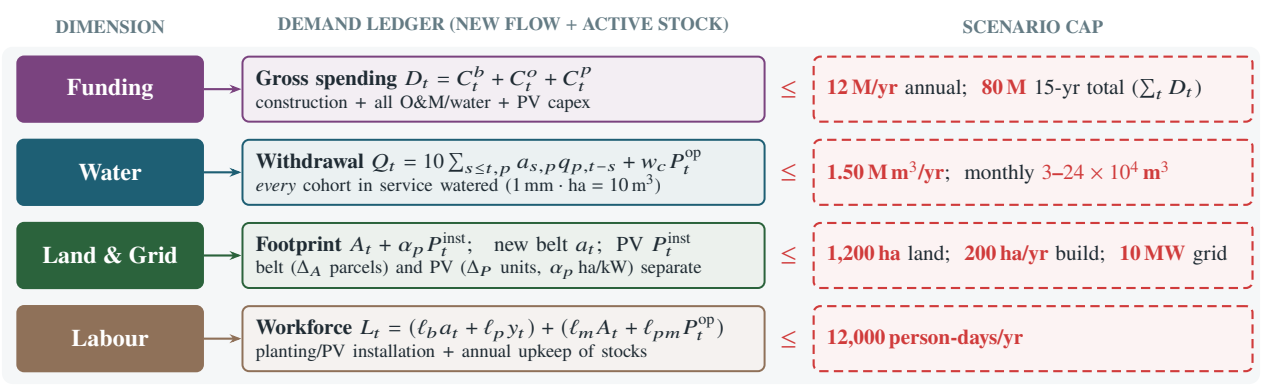

DIMENSION
DEMAND LEDGER (NEW FLOW + ACTIVE STOCK)
SCENARIO CAP
Funding
Gross spending $D_t = C_t^b + C_t^o + C_t^p$
construction + all O&M/water + PV capex
≤
12 M/yr annual; 80 M 15-yr total ($\sum_t D_t$)
Water
Withdrawal $Q_t = 10 \sum_{s \le t,p} a_{s,p} q_{p,t-s} + w_c P_t^{\text{op}}$
every cohort in service watered (1 mm · ha = 10 m³)
≤
1.50 M m³/yr; monthly 3–24 × 10⁴ m³
Land & Grid
Footprint $A_t + \alpha_p P_t^{\text{inst}}$; new belt a_t; PV P_t^{inst}
belt (Δ_A parcels) and PV (Δ_P units, α_p ha/kW) separate
≤
1,200 ha land; 200 ha/yr build; 10 MW grid
Labour
Workforce $L_t = (\ell_b a_t + \ell_p y_t) + (\ell_m A_t + \ell_{pm} P_t^{\text{op}})$
planting/PV installation + annual upkeep of stocks
≤
12,000 person-days/yr